

should be primarily bonds. But if the fund's liabilities are the PBO, then stocks might be a more appropriate inflation hedge in the long run because wage growth, like economic growth in general, is correlated with stock returns (see chapter 7). ERISA has caused many large plans to *de-risk* partly because it emphasizes the ABO. Milliman reports that the one hundred largest pension plans have reduced their equity allocation from 80% in 2005 to below 40% in 2011.⁴⁶ The ideal, from the beneficiaries' point of view, is to match the *factor* exposures of the assets and liabilities, so that the promised pensions can be met in most economic environments.

Management

Pension funds share many of the management problems of SWFs—even though pension funds in the private sector should theoretically be able to find competent people, fire incompetent people, and build optimal management structures.

The pay and skill of people working for pension funds resembles a pyramid. At the top of the pyramid, the board gets paid the least, and some board members often have little or no investment knowledge. Public pension plan boards, especially, can be dominated by politicians or union members with little financial expertise. In the middle of the pyramid sit the pension fund managers. Their pay often increases with the AUM they are responsible for, rather than the actual value they generate. The bottom of the pyramid is where most of the money is paid out in compensation—particularly to external managers in the form of fees.

The biggest decisions that affect the pension fund's portfolio resemble an inverse pyramid. The most important decision for the fund is how much risk to take and which factor risk premiums to collect.⁴⁷ This is a decision made by the board members—usually the least compensated and least informed parties involved. At the bottom of the pyramid, the highest paid fund managers trying to find elusive alpha (outperformance relative to a benchmark, see chapter 10) generate returns that have the smallest impact on the fund. They get paid the most, yet what they do matters least.

Public pension plans, and many private ones as well, are too often inefficient bureaucracies. They are hampered in what they can pay; the talent and expertise are on the wrong side, making them inviting prey for the predators of Wall Street. Building an organization of skilled investment professionals is hard. Compensation does matter. But most important is the creation of management structures that emphasize responsibility and accountability, where employees are measured against sufficiently rigorous benchmarks, and there are consequences for failing to meet those benchmarks. Some organizations have countered bureaucracy by creating independent management organizations. Canada Pension

⁴⁶ General Motors is a case study in de-risking. See "GM Asset Management and Martingale's Low Volatility Strategy," Columbia CaseWorks ID #110315.

⁴⁷ See chapters 14 and 15.